Thea: Towards the Harness of Embodied Agents

Revisited after GPT-6 Astra

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https://eit-hai.github.io/thea/

Abstract: THEA carried the harness paradigm of coding agents into the physical world, adding the two things the physical world does not supply for free: a persistent readable world state and a verifiable judgment of each action (Section 1). With GPT-6 Astra, a model can be handed a robot SDK inside an ordinary coding-agent environment. We revisit at what granularity it now controls the body (Section 2), who supplies each harness component, the model, the coding harness, or something still to be built (Section 3), and what changes and what remains (Section 4).

1 Thea: A Harness for the Physical World

Coding agents write, run, read the error, fix, and run again [1, 2]. What made them work is not a better model but a better *harness*: the infrastructure that grounds the model in its environment, mediates actions through tools, and closes the loop through evaluation and recovery [3, 4, 5]. THEA [6] carries this blueprint to robots (Figure 1): an agentic loop, every robot capability as a callable tool, context engineering, skills, memory, an Embodiment Profile, and hooks.

The physical world withholds two things software grants for free. A codebase is readable and persistent. A room is seen as a momentary slice, and the world keeps no record. A command returns an exit code. A closed gripper returns no verdict on whether the cup is in it. *Scene Graph as Context* turns local observations into persistent symbols briefed before every decision. *Evaluation as Exit Codes* runs an independent evaluator after each manipulation that judges success against the tool's post-condition and returns a failure reason for recovery. Its accuracy caps what retries can recover. Deployed on three robots with GPT-5.5 as the decision model, THEA outperformed end-to-end policies [7, 8, 9] and orchestration baselines [10, 11] built on the same tools. Every component sat on the harness side of the model–harness boundary. This abstract asks where that boundary sits two months later.

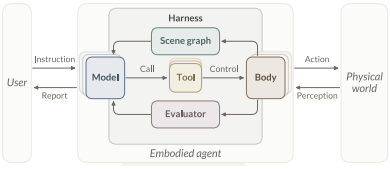


Figure 1: The THEA harness: tools, a persistent scene graph, and an independent evaluator [6].

2 Revisiting the Tools

In THEA the model chose among *skill-level* tools, `pick`, `place`, `navigate_to`, each a learned policy or stack [12]. GPT-6 Astra changes the premise: a frontier model can now be handed the robot itself [13]. In October 2026 we gave it, running in a coding-agent environment (Codex), the Astribot S1 Python SDK and its documentation, with no THEA component present, and asked for four tasks with the left parallel gripper (Figure 2(a)): tabletop cleanup, cleanup after a base approach, drawer retrieval and closure, and bowl transport (with a cube inside).

We observe the agent working *below* the pick-and-place interface: it registered head-camera RGB-D into the base frame, solved inverse kinematics numerically against the SDK's forward kinematics, screened joint samples for self-collision, sent one motion segment at a time, and drove the base with velocity pulses gated by lidar clearance. Its first errors were geometric and self-corrected, and all four tasks reached their final states. Within this scope, learned manipulation policies were not needed: the model wrote the control stack THEA had put behind a tool.

3 Revisiting the Harness

The capability on display is not the model's alone: Codex supplied the loop, tool execution, files and the write-run-read-fix cycle. What it does not supply is anything specific to the physical world. We therefore read the runs through the components THEA had built, asking which the model already has and which it still lacks.

World state and spatial memory. In LHPR-VLN [14] and SuperNav [15] without a *scene graph*, targets were re-localized from images and earlier sightings kept in conversation. With instructed graph memory, stored room and route records were reused (Figure 2(b)). Across three paired SuperNav tasks (one run per condition, 500-action cap, 1 m threshold), completion rose from 2/3 without the graph to 3/3 with instructed graph memory, and mean SPL from 0.485 to 0.690 (+42.3%). SPL weights task success by the shortest-to-actual path length ratio, with failures scored zero.

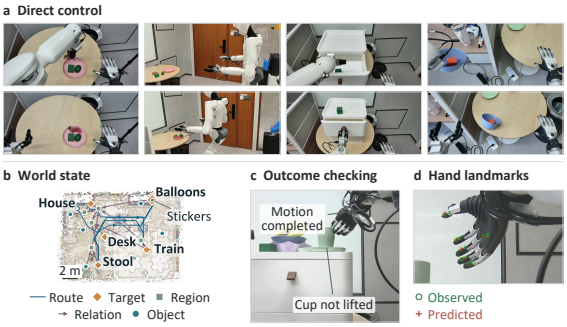


Figure 2: (a) Left parallel-gripper tasks, left to right: tabletop cleanup, cleanup after a base approach, drawer retrieval and closure, and bowl transport (with a cube inside). Execution above, outcomes below. (b) Recorded route and instructed scene graph over a scene map, with model-estimated node positions. (c) Revo2 reaches its motion target, but the cup stays on its support. (d) After slight finger closure, observed hand-surface landmarks (green rings) versus calibrated model predictions (red crosses). Click the blue panel labels for videos and experiment records.

Evaluation. We examine whether the model's own judgment of outcomes is enough (*evaluator*). In the Revo2 cup-grasping trials, SDK motion completion did not imply a successful grasp: the wrist rose while the cup stayed on its support (Figure 2(c)). The agent checked post-action images, rejected unsuccessful lifts and adjusted its grasp. These were task-specific checks of unmeasured accuracy, not an independent evaluator.

Body limits. We examine whether the model adapts to a new body and its constraints (*embodiment profile*). The agent found that the installed collision model still described a parallel gripper rather than the Revo2 hand. Human-prompted geometry checks gave local per-finger landmark RMS errors of 1.35–2.20 mm with a fixed hand shape, versus 5.26 mm overall after slight finger closure (Figure 2(d)). Calibration for one hand shape did not guarantee accuracy after closure.

Skills. We examine whether experience becomes reusable skills (*skills and memory*). Modules written for the first task (capture, localization, IK, screening) were reused in the other three, accumulating as files in the workspace.

4 What Changes and What Remains

After Astra, we re-examine the harness for what has changed and what has not. The two gaps have not: the runs still had to re-sense the world and judge their own outcomes. What changes is the responsibility the harness carries around them. *Tool granularity blurs downward.* THEA's tools were policies. Here the model wrote motion segments against the SDK, so the harness must expose the body at several levels. *The Embodiment Profile does more work.* When the model emits poses directly, kinematic limits, workspace and calibration are no longer hidden inside a policy, and the profile becomes the layer on which zero-shot cross-embodiment rests. *Safety moves into the harness.* A verified grasp policy carries part of its own safety. A raw pose command carries none, so permissions and sandboxes become workspace bounds, collision checks and force limits, and the lower the abstraction, the heavier this layer. *The evaluator is the precondition for self-improvement in the physical world.* Digital self-improvement ran first because exit codes and unit tests existed [13]. The physical world has none, so Evaluation as Exit Codes must come first.

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